EARTH LAB / SCIENTIFIC PYTHON

SpectralBridge

From flightlines to analysis-ready data

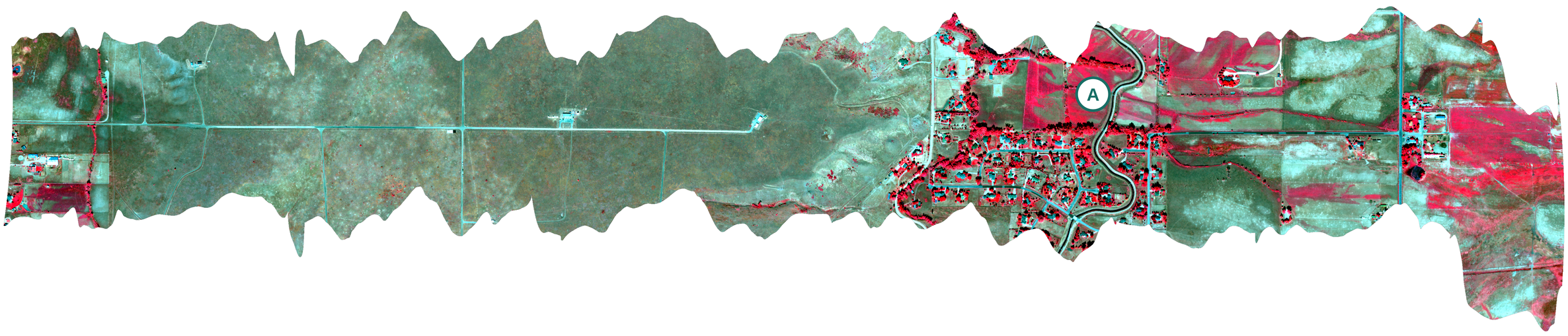
Prepare and compare reflected light across airborne, drone and sensor-like products.

REAL FLIGHTLINE / VISIBLE LIGHT

A flightline is the strip recorded along one flight path.

SAME GROUND / INFRARED SHOWN IN RED

A marks the same patch in both images.



Infrared reveals contrasts our eyes miss. Red is a display color, not heat or fire.

Illustration: two views of one corrected image, not a before-and-after correction test.

What the package does

The NEON workflow turns a spectral image into reusable scientific products.

01 Prepare flightlines

Download NEON data and export spectral images.
Use a separate route for supported local drone files.

Readable images + metadata

02 Correct viewing effects

Account for terrain and viewing geometry
with BRDF and topographic correction.

Corrected spectral images

03 Make sensor-like products

Combine wavelengths using configured sensor bands,
including Landsat and MicaSense representations.

Configured target-band images

04 Export and check

Extract pixel or polygon tables, merge outputs,
and inspect quality plots, metrics and reports.

Parquet tables + quality evidence

Choose the workflow that fits your data

NEON flightlines
Process one flight or a collection.
`go_forth_and_multiply`

Local drone imagery
Process supported TIFF/HDF5.
`run_drone_pipeline`

Completed products
Compare flights with compact statistics.
`run_bulk_pipeline`

Spectral libraries
Inspect inputs and report patterns.
`run_spectral_library_analysis`

Built for long runs Reuse valid outputs. Work in chunks. Keep files and provenance traceable.

Sensor-like products support comparison; agreement with independent observations still needs validation.

Explore the documentation
earthlab.github.io/spectralbridge

github.com/earthlab/spectralbridge

Image: NEON R10C, 15 September 2021 | Native 1 m pixels | Original footprint in both views

Display only: per-channel 2-98% contrast stretch, gamma 0.8. White outside the flightline means no image data.

Visible RGB near 650/550/470 nm; infrared RGB near 850/650/550 nm | Package source and local docs | 27 October 2026